

Secure Your Success: Simplifying Cybersecurity for Small Businesses

CYBERSECURITY FOR

SMALL BUSINESS



September 20, 2025

Who am I?

Identity Paradox

How do you know
(I'm not a deep-fake)?



Who am I?

Ron Woerner

- Hacker
- CyberSecurity Consultant / Trusted Advisor
- Professor, Bellevue University
- 25+ years experience in IT / Security
- Blogger, writer, and podcaster

Slides available at <https://github.com/hackerron/SMB-Cyber/>



Websites & Social Media:
<https://linktr.ee/cyberron>



LinkedIn:
<https://www.linkedin.com/in/ronwoerner/>



WHY

are we here?

Agenda

Secure Your Success: Simplifying Cybersecurity for Small Businesses

Why should I care?

What's going on?

Attack methods / Top issues

Your responsibility

Where to go from here

You'll learn:

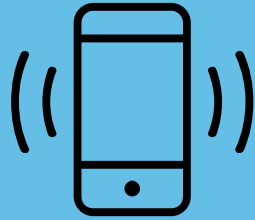
- How to protect your business from common cyber threats
- Best practices for securing your digital assets and customer data
- How to recognize and respond to phishing, scams, and other online risks
- The basics of artificial intelligence (AI) Security and how it can benefit your business
- Practical steps you can take today to improve your cybersecurity posture

By the end of this session, you'll walk away with a clearer understanding of how to safeguard your business online and how to leverage AI responsibly and effectively.

During this session



Ask questions if
you're unsure or
need help.



Please limit
other
distractions.



Stay focused on
the session's
outcomes.



Have fun!



Stay open minded
and tell us what you
REALLY think.

Psychological Safety

This means...

- Postpone / withhold your judgement of IDEAS
- Encourage WILD exaggerated ideas
- Quantity counts at this stage NOT quality
- Build on IDEAS put forward by others
- EVERY person and every idea has worth.



What is your biggest security concern?

1. Organization Data Breach
2. Identity Theft
3. Ransomware
4. Malware/Viruses
5. Emerging Technologies
6. “Unknown unknowns”
7. Other



Consequence:

- Reputation
- Revenue
- Regulation

What About You?

What are you doing to
reduce your cyber risks?

Who is using AI?

Billions Lost!



Black Hat Hackers



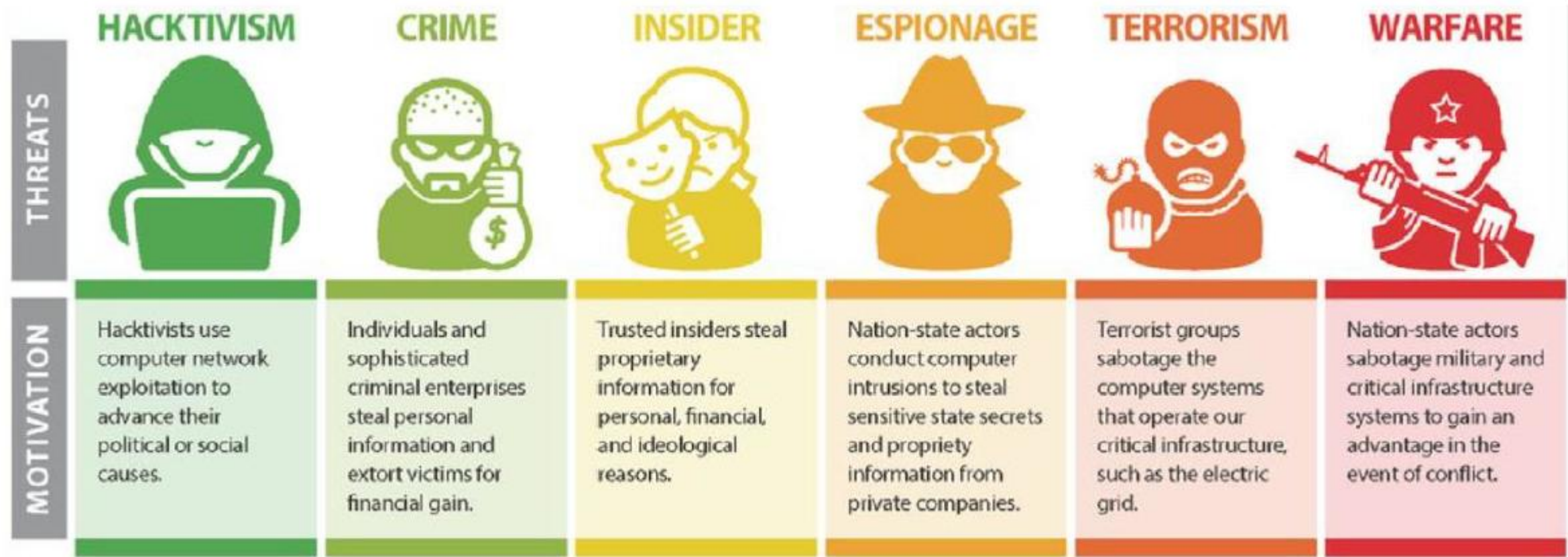
Cyber War!

Nation-State Threats:

- Russia
- China
- North Korea
- Iran

Types of Cyber Threats

Figure 1: Cyberthreat Spectrum



Why Cyber Safety?

Phishing

30%

of breaches were linked to third-party involvement, twice as much as last year, and driven in part by vulnerability exploitation and business interruptions

54%

of perimeter-device vulnerabilities were fully remediated by organizations in the past year, while almost half remained unresolved

34%

increase in attackers exploiting vulnerabilities to gain initial access and cause security breaches compared to last year's report

44%

of all breaches analyzed showed ransomware was present, marking a notable rise from last year's report

Ransomware

Source: 2025 Verizon Data Breach Report - <https://www.verizon.com/business/resources/reports/dbir/>

Social Engineering

Cybersecurity for SMB – September 2025

Personal Information



Who you are

Where you are

What you are doing

Recent Scams

 **Better Business Bureau, Inc.**
567 followers
3h • 🌐

#WednesdayWarning 📱 Smishinggg (text messaging) scams are rampant during the COVID-19 crisis. 📧 Scammers are using everything in their arsenal to try and trick you to not think before you click on that link. **DON'T CLICK THE LINK!** 💡 Delete the text message right away. **#WisdomWednesday #StartWithTrust #Smishing #TextMessagingScam #ScamAlert** <https://lnkd.in/eJBbEdU>

Text Message
Yesterday 10:50 AM

██████████, we found a package from March pending for you. Kindly assume ownership and confirm for delivery here: l5ssv.info/QTGGdgoLks

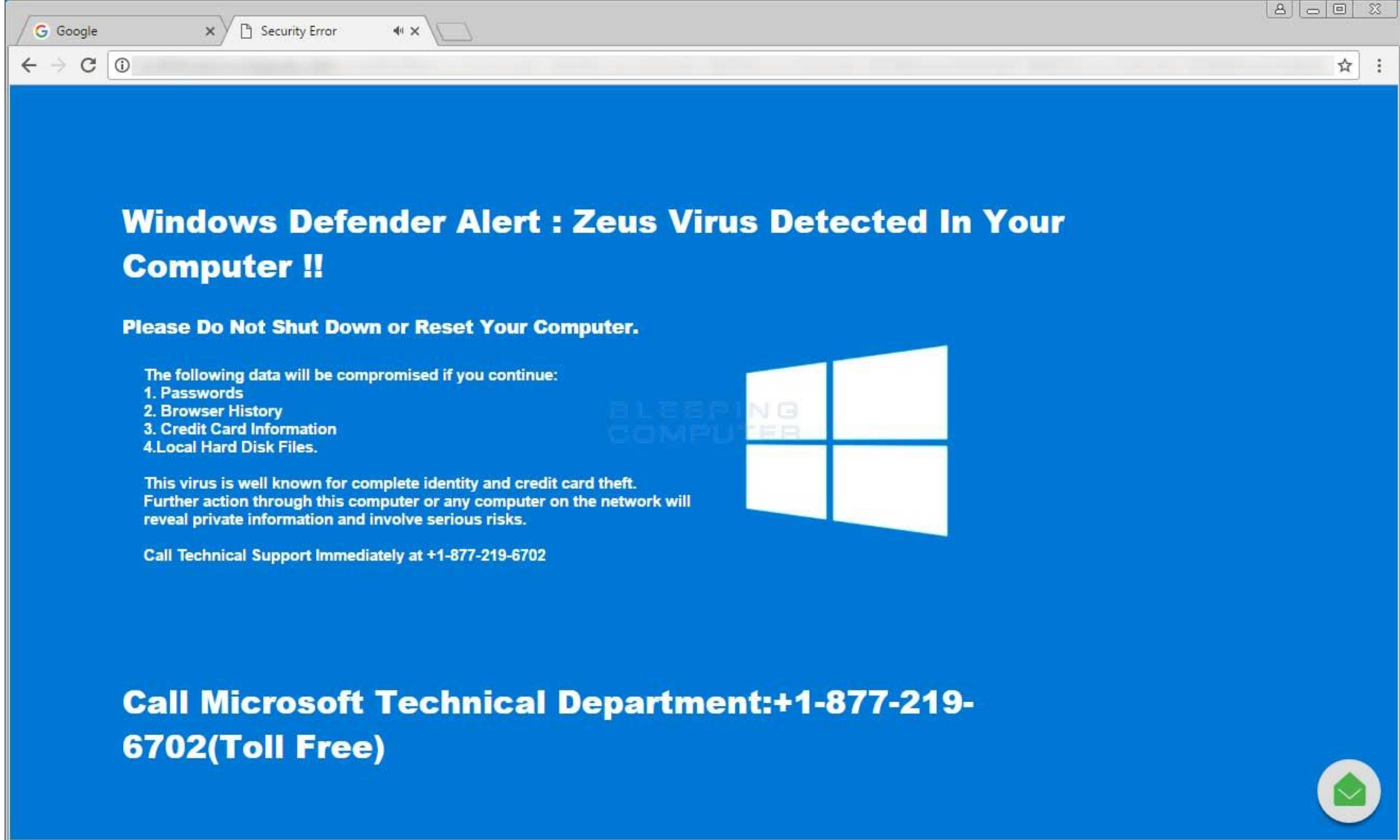
KETV.COM

Receive a text message about a pending package? Don't click the link — it's a scam, officials say

When you need help online, a scammer will come to the rescue



<https://www.linkedin.com/pulse/fake-bookingcom-elaborate-info-stealing-campaign-david-sehyeon-baek/>
<https://brothke.medium.com/when-booking-com-becomes-booking-scam-9ae605a12790>



Cyber Attacks and Threats

- Social Engineering / Human Hacking / Phishing
- Business Email Compromise (BEC) / Account Takeover
- Ransomware
- Stolen Data
- Denial of Service - DOS/DDOS
- Third-party / Supply Chain
- Internal Threats



How they find victims

OSInt

Linked 



ChatGPT

 facebook

 hunter

Google



theHarvester

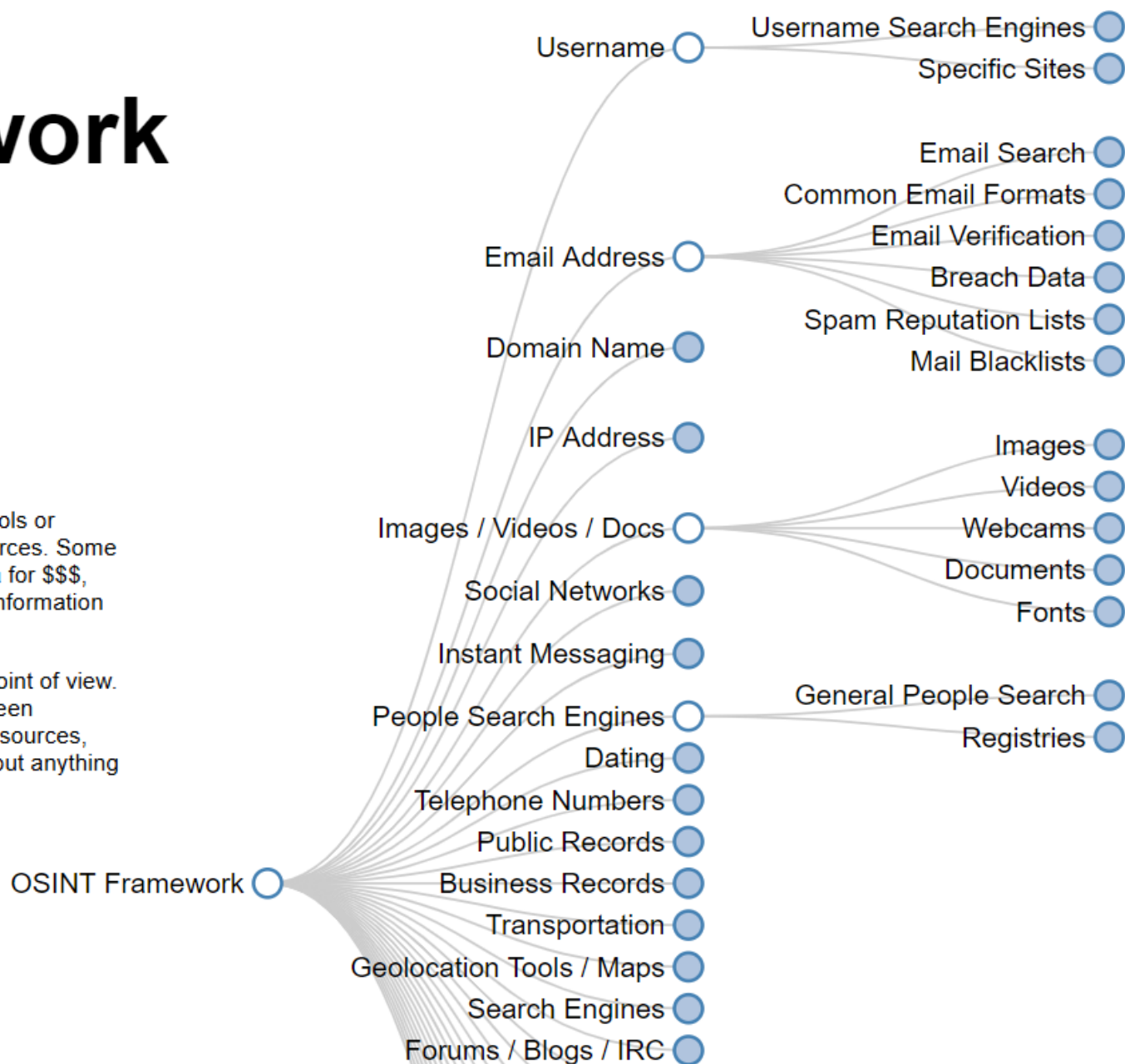
OSINT Framework

<https://osintframework.com/>

Notes

OSINT framework focused on gathering information from free tools or resources. The intention is to help people find free OSINT resources. Some of the sites included might require registration or offer more data for \$\$\$, but you should be able to get at least a portion of the available information for no cost.

I originally created this framework with an information security point of view. Since then, the response from other fields and disciplines has been incredible. I would love to be able to include any other OSINT resources, especially from fields outside of infosec. Please let me know about anything that might be missing!



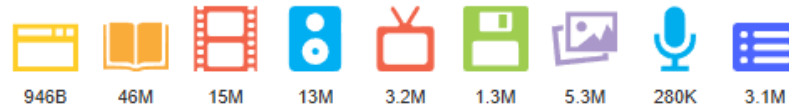
Search the history of over 946 billion web pages on the Internet.

WayBackMachine

Enter URL or keywords



Internet Archive is a non-profit library of millions of free texts, movies, software, music, websites, and more.



Search

GO

[Advanced Search](#)

Archive News

Calling All Musicians: Mini Concerts at the Internet Archive

Vanishing Culture: Why Preserve Flash?

Tom Lehrer (1928–2025): A Life in Satire, A Legacy in the Commons

[More posts](#)

<https://archive.org/>

Have I Been

Check if your email address is in a data breach

rwoerner@bellevue.edu|



Check

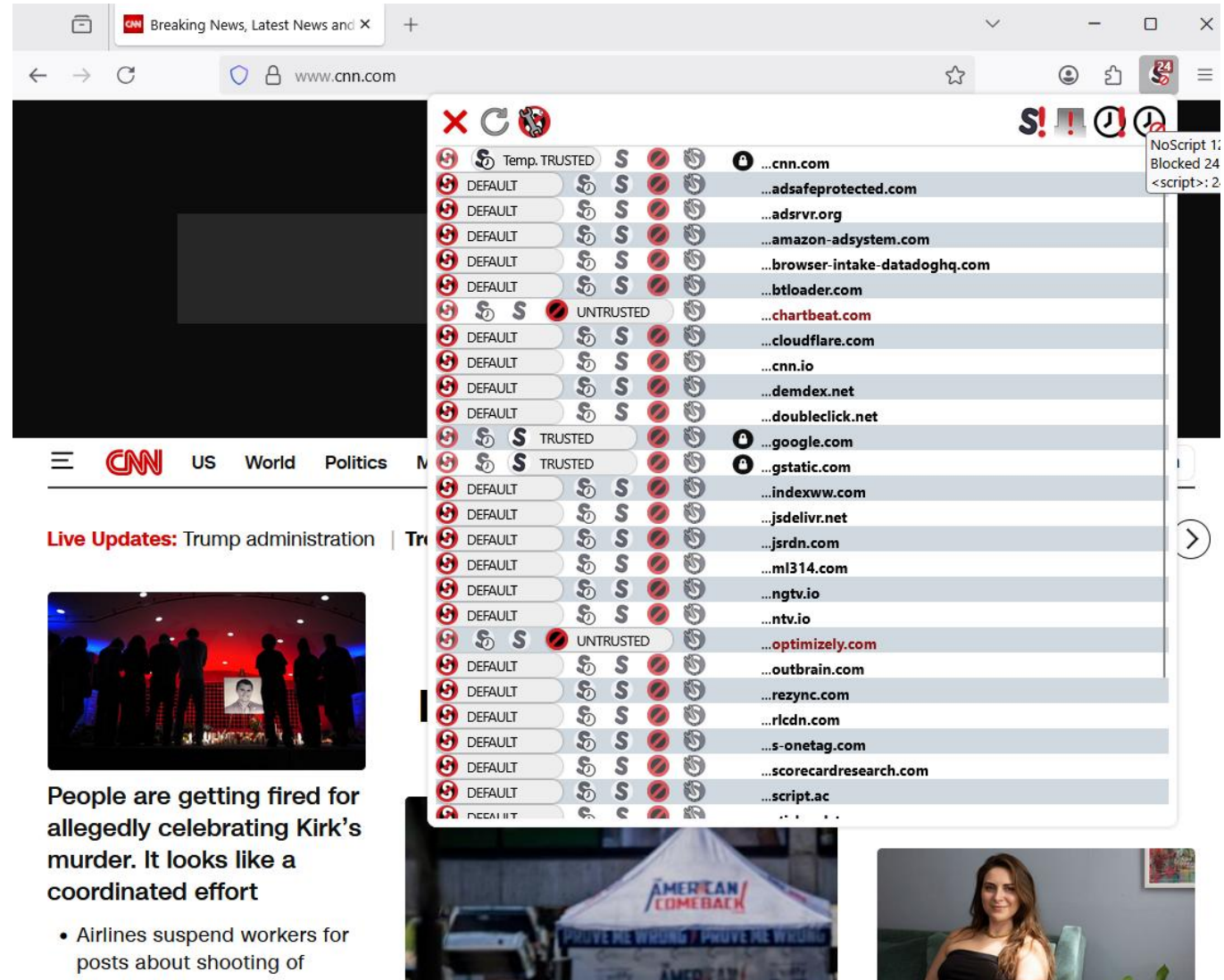
Using Have I Been Pwned is subject to the [terms of use](#)

<https://haveibeenpwned.com/>

Email Breach History

Timeline of data breaches affecting your email address

Website Tracking Demo



<https://noscript.net/>

Social Engineering - The Old Con Job

Preys on qualities of human nature:

- The desire to be helpful
- The tendency to trust people
- The fear of getting into trouble
- Acting without thinking

Technology is only a tool for manipulation



Blind Trust

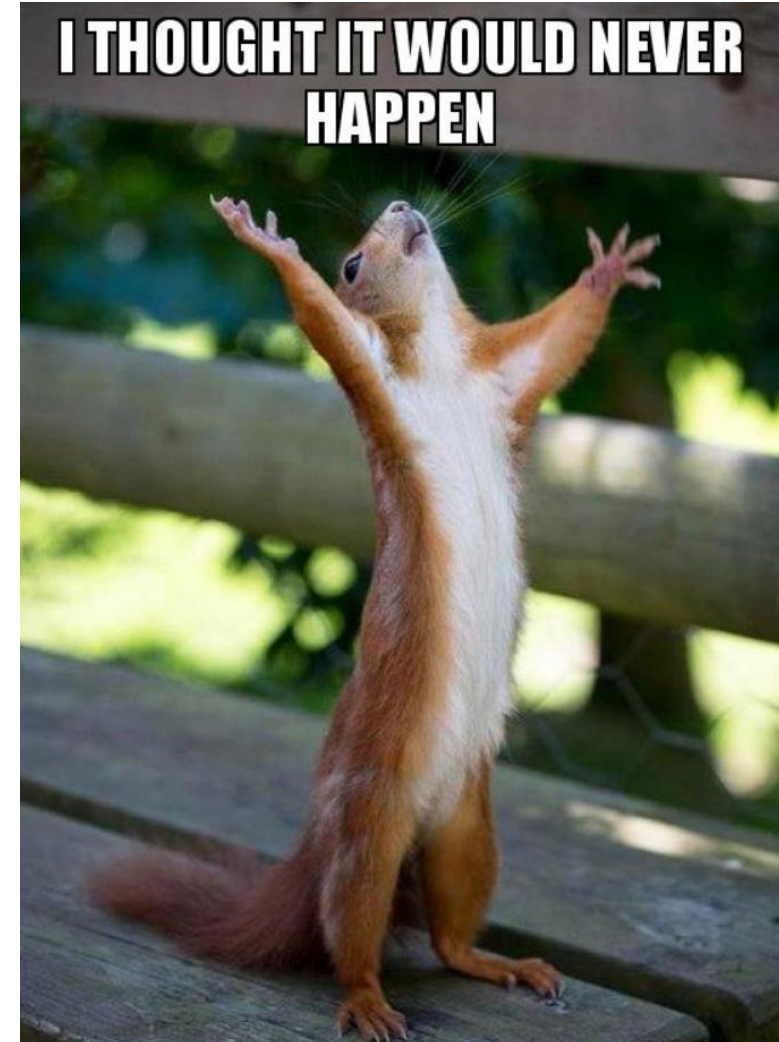


“Just because you see something on the Internet with a quote, a picture and a date, it doesn’t mean it’s going to be true.”

- Abraham Lincoln, 2006

Trust, *and* Verify

Complacency



Mindlessness



Busy-ness



Advanced Persistent Threats (APTs)



Dark Web vs. Deep Web



Dark Web vs. Deep Web

Deep Web: Any part of the internet that is **not indexed by search engines**. It includes content that requires authentication or special access

Examples:

- Password-protected websites: Areas of the web that you can only access with valid login credentials.
- Webmail services: Your email inbox, which is private and requires a login.
- Private databases: Information stored behind security measures.

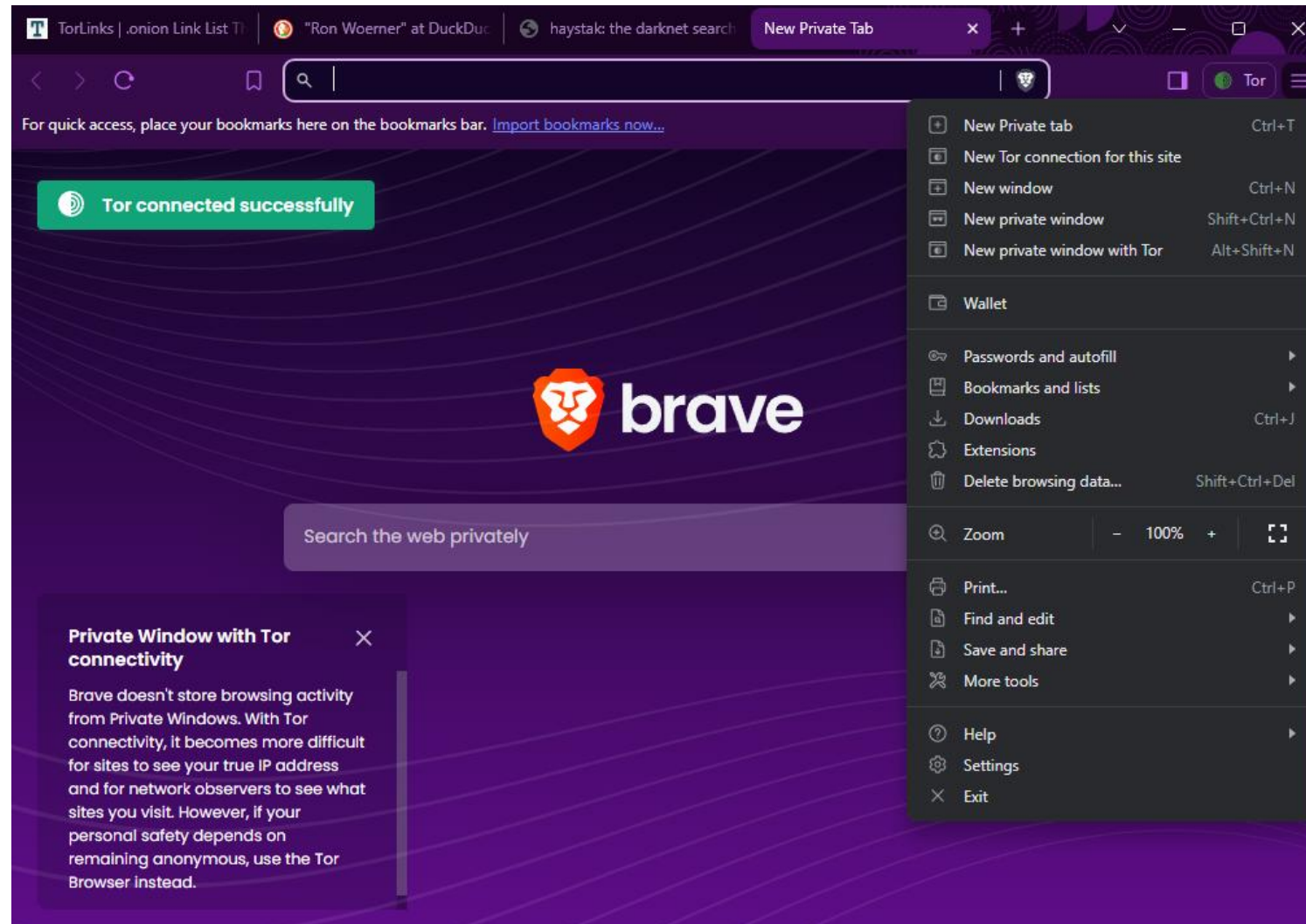
Accessible via regular web browsers, but you need specific permissions to view its content.

Dark Web: A subset of the deep web that can only be accessed using special tools, such as the Tor network.

- Anonymity: Users on the dark web remain anonymous due to encryption and routing through multiple servers.
- Content:
 - Illegal activities: The dark web is known for hosting illegal marketplaces, drug trafficking, and other illicit services.
 - Whistleblower platforms: Some use it to share sensitive information anonymously.
- Not indexed by search engines, making it intentionally hidden from public view

Accessing the Dark Web

***** USE AT YOUR OWN RISK!!!**



AI Risks & Threats

Manipulating victims:

- Pissed off,
- Perturbed, or
- Panicked

Prompt: I'm building a presentation on cybersecurity and AI for a technical audience. Provide 5 ways ChatGPT and AI can be used maliciously.

- Automated social engineering
- Supercharged phishing
- Deepfakes and disinformation
- Enhanced cyber reconnaissance
- Malware automation & mutation

**The first principle
is that you must
not fool yourself
and you are the
easiest person
to fool.**

Richard P. Feynman

THN The Hacker News

Cursor AI Code Editor Flaw Enables Silent Code Execution via Malicious Repositories

A security weakness has been disclosed in the artificial intelligence (AI)-powered code editor Cursor that could trigger code execution when...

2 days ago



Cybersecurity Dive

AI is helping hackers automate and customize cyberattacks

CrowdStrike's annual cyber-threat-hunting report reveals the double threat that AI poses to many businesses.

1 month ago



THN The Hacker News

⚡ Weekly Recap: Drift Breach Chaos, Zero-Days Active, Patch Warnings, Smarter Threats & More

Cybersecurity never slows down. Every week brings new threats, new vulnerabilities, and new lessons for defenders. For security and IT teams...

6 days ago



THN The Hacker News

You Didn't Get Phished — You Onboarded the Attacker

North Korean hiring fraud surged 220% in 2023, using AI deepfakes to infiltrate companies and steal data.

6 days ago



THN The Hacker News

[Webinar] Shadow AI Agents Multiply Fast — Learn How to Detect and Control Them

... ⚠️ One click is all it takes. An engineer spins up an "experimental" AI Agent to test a workflow. A business unit connects to automate...

5 days ago



THN The Hacker News

Threat Actors Weaponize HexStrike AI to Exploit Citrix Flaws Within a Week of Disclosure

Threat actors are attempting to leverage a newly released artificial intelligence (AI) offensive security tool called HexStrike AI to...

1 week ago



AI Risks & Threats

NIST AI 100-1

AI RMF 1.0



Fig. 1. Examples of potential harms related to AI systems. Trustworthy AI systems and their responsible use can mitigate negative risks and contribute to benefits for people, organizations, and ecosystems.

Malicious AI Examples

Automated Social Engineering:

- **Problem:** AI-powered chatbots can engage in seemingly natural conversations, impersonating customer service representatives or trusted individuals to extract sensitive information.
- **Impact:** Increased risk of identity theft, financial fraud, and data breaches by tricking users into revealing personal details.
- **Example:** Imagine an AI chatbot posing as a bank representative calling you to "verify" your account information, eventually luring you into disclosing your PIN or credentials.

Supercharged Phishing:

- **Problem:** ChatGPT excels at mimicking human writing styles and crafting personalized narratives. This makes AI-generated phishing emails more convincing and bypasses traditional spam filters.
- **Impact:** Increased risk of sensitive data breaches, financial losses, and reputational damage for organizations.
- **Example:** Imagine an email, Teams, Slack, etc. message seemingly from a trusted colleague praising your recent work and subtly prompting you to click a malicious link to access a "bonus document."

Deepfakes and Disinformation:

- **Problem:** AI can be used to manipulate audio and video to create realistic deepfakes that spread misinformation, damage reputations, and sow discord.
- **Impact:** Erode trust in institutions, incite social unrest, and manipulate public opinion for nefarious purposes.
- **Example:** A fabricated video portraying a political leader making inflammatory statements could go viral and disrupt democratic processes.

Malicious AI Examples

Enhanced Cyber Reconnaissance:

- **Problem:** AI can analyze vast amounts of data to identify vulnerabilities in networks, systems, and software, aiding attackers in targeting their efforts..
- **Impact:** Increased risk of successful cyberattacks as attackers gain valuable insights into potential entry points and exploit weaknesses.
- **Example:** Imagine an AI scouring open-source forums to find disgruntled employees mentioning security protocols, providing valuable intel for targeted attacks.

<https://osintframework.com/>

Malware Automation and Mutation:

- **Problem:** AI can be used to automate the creation and modification of malware, making it more sophisticated, evasive, and difficult to detect by traditional security measures.
- **Impact:** Increased risk of widespread malware outbreaks, data loss, and disruption of critical infrastructure.
- **Example:** Imagine an AI generating constantly evolving variants of a ransomware virus, making it nearly impossible to identify and neutralize before causing widespread damage.

WormGPT, <https://flowgpt.com/p/wormgpt-6>

What You Can (Should) Do

**Only You can
Protect Yourself
and Others**



**Security and Privacy
Is
Your
Responsibility**

Pathway to Cybersecurity

Starts with BOD and Executive Committee

Governance

- Data
- AI
- Privacy
- Enterprise Risk
- Supply Chain & Third Party
- Information Security
- IT

Strategies should exist that compliment and build upon each other

Management Strategies & Programs

- Enterprise Risk Management
- Data Privacy Management
- AI
- Information Security Management
- Project Management

Advice for Businesses: Respect Privacy

- Know your data / PII
- If you collect it, protect it
- Consider adopting a privacy framework
- Conduct an assessment
- Transparency builds trust
- Maintain oversight (Self and partners/vendors)



Source: <https://staysafeonline.org/data-privacy-day/about-dpd/>

Maturing Your Cyber Security Program

- Culture of Security
- Governance & Compliance
 - External (Regulations & Cyber Defense Frameworks)
 - Internal (Policies, Standards, Procedures)
- Risk Management / Economics
- Supply Chain & Third Party
- IT Architecture
- Intrusion Detection and Response
- Security Awareness

Simple Steps for Cyber Security

- “Trust, *and* verify”
Be a little skeptical
Critical for AI
- Account Security
Multi-factor authentication
Password keepers
- If you see something,
say something
- “Be Prepared”
It’s not if, but when...
- Update your PC & apps
Pay a little now or a lot later

When in doubt, check it out (ask)

Multi Factor Authentication



⚠ Archived Content

In an effort to keep CISA.gov current, the archive contains outdated information that may not reflect current policy or programs.

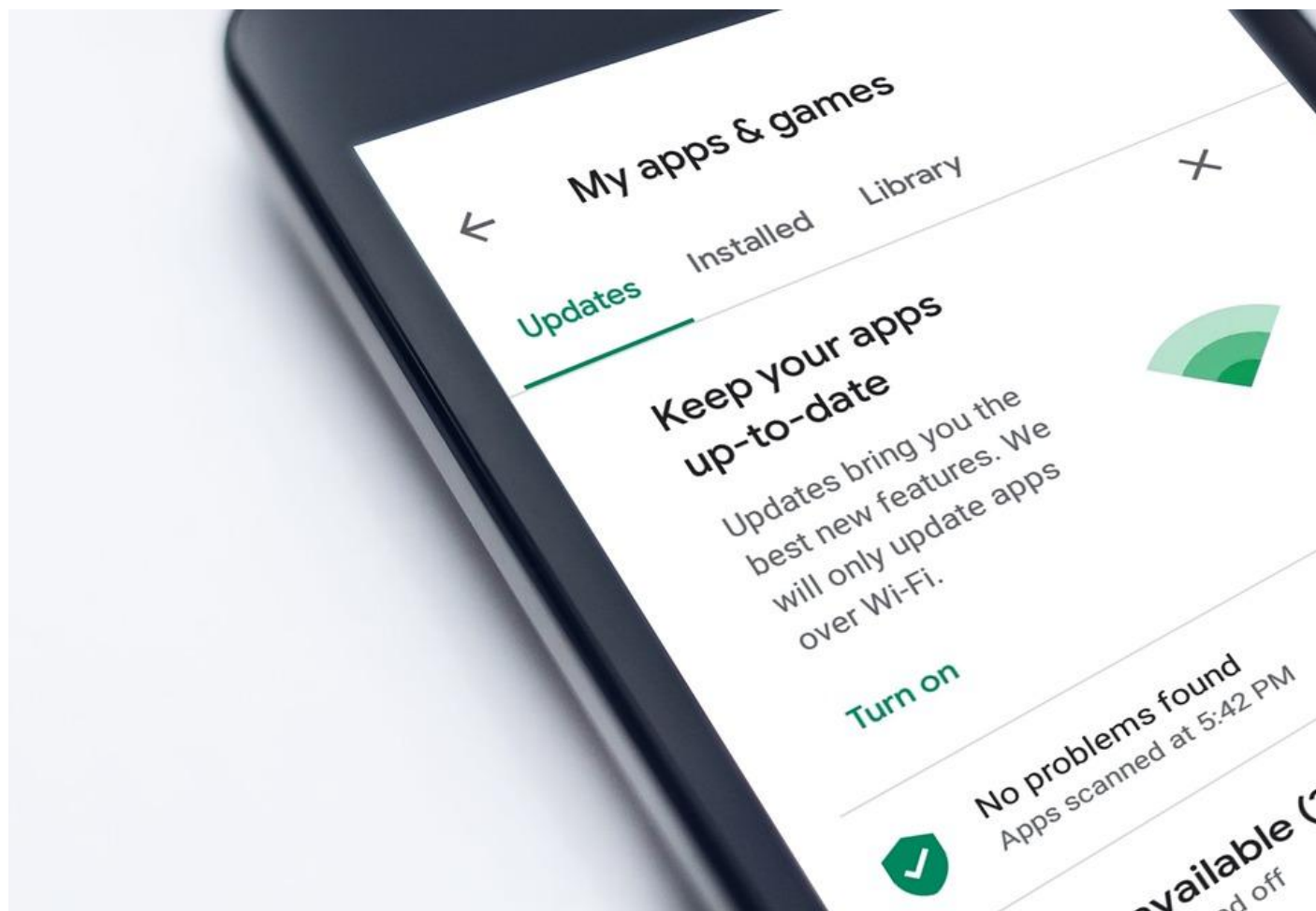
<https://www.cisa.gov/MFA>

Malware = Malicious Software



<https://www.virustotal.com/>

Stay Up to Date



Ransomware

Payment information

90 BTC

UNPAID

Send 90 BTC (in ONE payment) to:
don't include transaction fee in this amount

17cK4w6E4apxCQSmu5p7Pvk3zBGoEsuQgH

Check payment

Available once every 12 hours



RaaS -- Ransomware-as-a-Service
[ProLock](#), [NetWalker](#), & [LokiBot](#)

<https://www.nomoreransom.org>

<https://www.cisa.gov/stopransomware>

Incident Response

- Set your Response Strategy & Plan
<https://frsecure.com/incident-response-plan-template/>
- Detecting incidents
- Incident analysis
- Response
- Checklists & Documentation
- Testing & Training

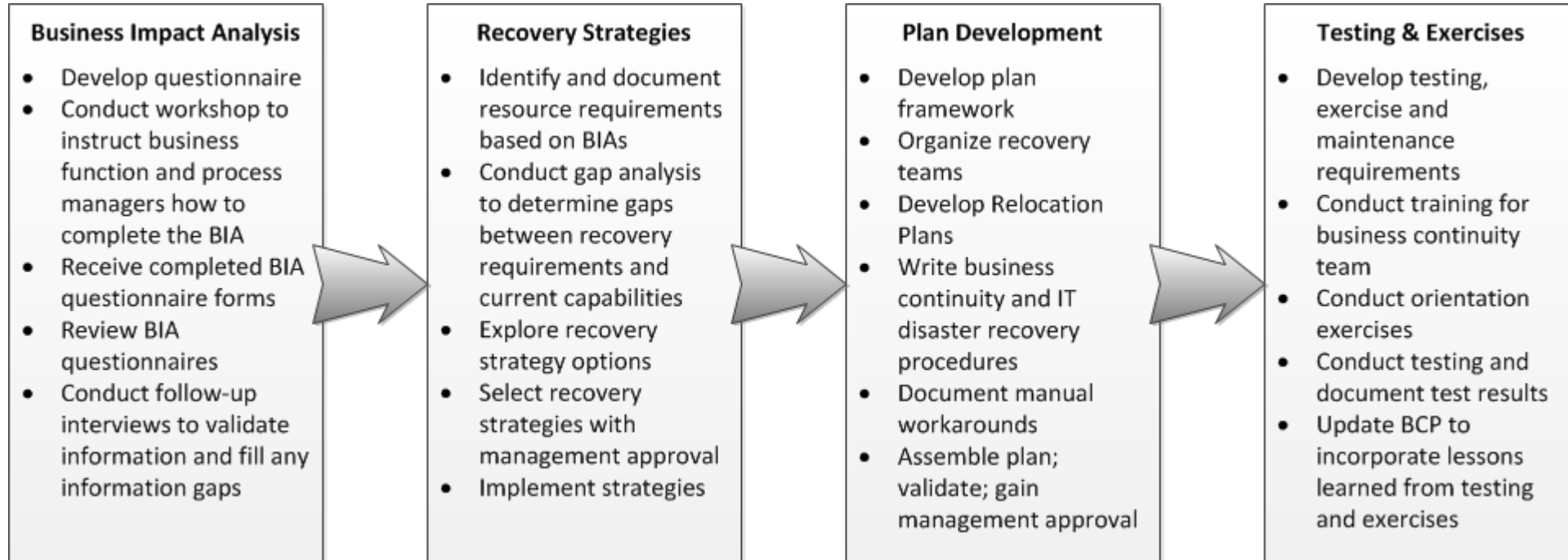


<https://www.cisa.gov/stopransomware>

Going for help

- Incident Response Plan
- Tech support
- Legal & Insurance
- Frauds & Scams
 - FBI, Internet Crimes Complaint Center (IC3): <https://www.ic3.gov/>
 - CISA Report: <https://myservices.cisa.gov/irf>
 - Cybercrime Support Center: <https://fightcybercrime.org/>
 - BBB Scam Tracker: <https://www.bbb.org/scamtracker/>

Business Continuity Planning



DHS – [Ready.Gov](https://www.ready.gov)

CISA SMB Resources



America's Cyber Defense Agency

NATIONAL COORDINATOR FOR CRITICAL INFRASTRUCTURE SECURITY AND RESILIENCE

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<https://www.cisa.gov/cyber-guidance-small-businesses>

Cyber Guidance for Small Businesses



A Different Kind of Cybersecurity Advice

Small businesses often do not have the resources to defend against devastating cyber threats like ransomware. As a small business owner, you have likely come across security advice that is out of date or that does not help prevent the most common compromises. For example, odds are that you have heard advice to never shop online using a coffee shop's wi-fi connection. While there was some truth to this fear a decade ago, that's not how people and organizations are compromised today. The security landscape has changed, and our advice needs to evolve with it.

This advice is different.

Below, we offer an action plan informed by the way cyberattacks actually happen. We break the tasks down by role, starting with the Chief Executive Officer (CEO). We then detail tasks for a Security Program Manager and the Information Technology (IT) team. While following this advice is not a guarantee you will never have a security incident, it does lay the groundwork for building an effective security program.

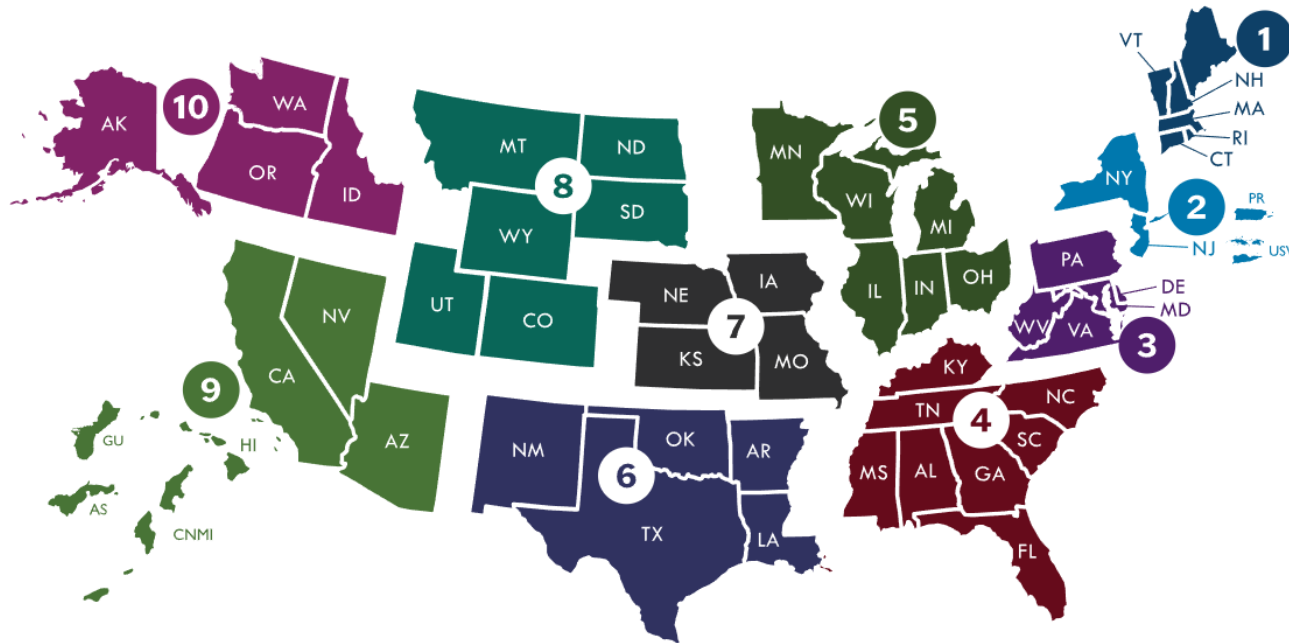
** This page was updated in April 2024.*

More CISA Resources



<https://www.cisa.gov/secure-our-world>

CISA Regional Offices



<https://www.cisa.gov/about/regions>

Cybersecurity Best Practices

<https://www.cisa.gov/resources-tools/services>

Artificial Intelligence

Cybersecurity for K-12 Education

Cybersecurity Awareness Month

Federal Cyber Defense Skilling Academy

Holiday Online Shopping

More than a Password

Open Source Software Security

President's Cup Cybersecurity Competition

Secure by Design

Organizations and Cyber Safety

Identify Theft and Personal Cyber Threats

Zero Trust

October – National Cybersecurity Awareness Month



<https://www.staysafeonline.org/cybersecurity-awareness-month>

2025 Theme and Messages

Every October, Cybersecurity Awareness Month teaches people simple ways to protect themselves online. This year's theme, **Stay Safe Online**, is all about the simple ways to protect yourself, your family and your business from online threats.

Small actions can make a big difference. That's why we're focusing on the **Core 4**. Four easy steps anyone can take to boost their online safety:

- Use strong passwords and a password manager
- Turn on multifactor authentication
- Recognize and report scams
- Update your software



<https://www.cisa.gov/cybersecurity-awareness-month>

SMALL BUSINESS CYBERSECURITY CORNER

<https://www.nist.gov/itl/smallbusinesscyber>

Cybersecurity Basics

NIST Cybersecurity Framework

Quick Start Guides

Events

Guidance by Sector

Guidance by Topic

Training

Videos

Join a Community of Interest

About

Cybersecurity @ NIST

+

+

+



SPOTLIGHT

CYBERSECURITY FOR SMALL BUSINESS

Source: <https://www.ftc.gov/tips-advice/business-center/small-businesses/cybersecurity>

 **FEDERAL TRADE COMMISSION**
PROTECTING AMERICA'S CONSUMERS

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TAGS: Consumer Protection | Privacy and Security | Data Security | Small Business

[Vea esta página en español](#)

CYBERSECURITY FOR SMALL BUSINESS

PROTECT YOUR SMALL BUSINESS

Learn the basics for protecting your business from cyber attacks. The business cybersecurity resources in this section were developed in partnership with the National Institute of Standards and Technology, the U.S. Small Business Administration, and the Department of Homeland Security.

 Cybersecurity Basics	 Understanding the NIST Cybersecurity Framework	 Physical Security
 Ransomware	 Phishing	 Business Email Imposters
 Tech Support Scams	 Vendor Security	 Cyber Insurance
 Email Authentication	 Hiring a Web Host	 Secure Remote Access

Securing AI for SMBs

NIST NATIONAL INSTITUTE OF
STANDARDS AND TECHNOLOGY
U.S. DEPARTMENT OF COMMERCE

Trustworthy & Responsible AI Resource Center

NIST AI Risk Management Framework

AI Risks & Trustworthiness

1. Valid and Reliable
2. Safe
3. Secure and Resilient
4. Accountable and Transparent
5. Explainable and Interpretable
6. Privacy-Enhanced
7. Fair – with Harmful Bias Managed

<https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf>

Securing AI for SMBs

AI for small business

Small businesses can use AI tools and applications to find solutions for all kinds of issues. Read up on both the risks and benefits.

Content

[How AI can benefit your small business](#)

[Risks of AI use](#)

[Common terms](#)

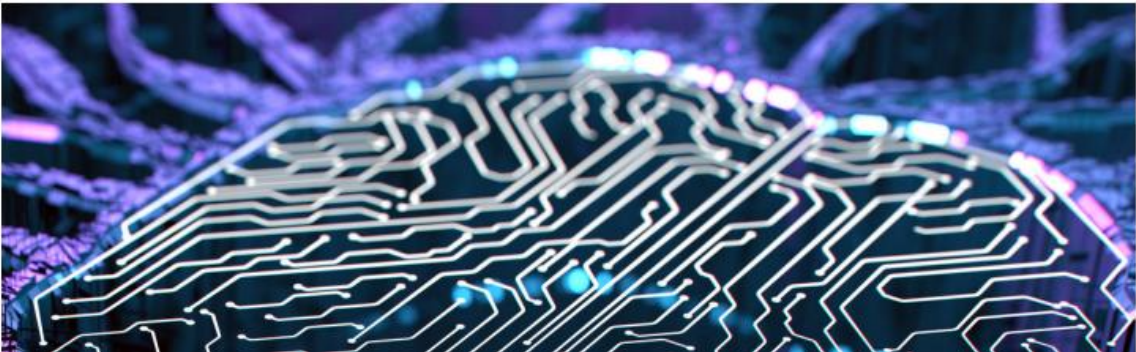
[Additional resources](#)

<https://www.sba.gov/business-guide/manage-your-business/ai-small-business>



Gemini can make mistakes, so double-check it

Guidelines for secure AI system development



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Who is responsible for developing secure AI?.....	7
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2. Secure development.....	12
3. Secure deployment.....	14
4. Secure operation and maintenance.....	16
Further reading.....	17



Guidelines for Secure AI Development,
<https://www.ncsc.gov.uk/files/Guidelines-for-secure-AI-system-development.pdf>,
NCSC (UK) & CISA (US)



The Obligatory Questions Slide

1 - aka Let's Play "Stump the Professor"

2 - Obligatory AI art

Apply What You've Learned

In the next week:

- Review the slide deck
- Pick 2-3 references for further learning

In the next month:

- Review the slide deck
- Pick 2-3 other references for further learning
- Teach at least 1 other person what you've learned



Resources

- DHS CISA: <https://www.cisa.gov/secure-our-world/secure-your-business>
- NIST Small Business Cybersecurity Corner: <https://www.nist.gov/itl/smallbusinesscyber>
- FTC: Cybersecurity for Small Business: <https://www.ftc.gov/tips-advice/business-center/small-businesses/cybersecurity>
- SBA: <https://www.sba.gov/managing-business/cybersecurity>
- StaySafeOnline (NCSA): <http://staysafeonline.org/>

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- How to recognize and respond to phishing, scams, and other online risks
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- Practical steps you can take today to improve your cybersecurity posture

By the end of this session, you'll walk away with a clearer understanding of how to safeguard your business online and how to leverage AI responsibly and effectively.

Thank You!

Ron Woerner



Ron Woerner

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<https://linktr.ee/cyberron>

